Lab: Data Cleaning and Preparation

Pre-Task Guidelines:

- Before beginning the tasks, please adhere to these guidelines.
- Create a workbook titled BI_Analyst_Capstone_ML (follow the format of M01L01) and save it.
- Import all CSV files into this workbook, allocating each CSV to its own worksheet.
- Ensure that you name each worksheet to correspond with the file names.
- Complete all the tasks outlined in the assignment overview for each CSV file, except for store_cities.csv and additional files.

Data:

Use the provided files - *sales.csv*, *product_hierarchy.csv*, *store_cities.csv*, *store_names.csv*, *city_names.csv*, and *product_names.csv*.

Hold the CTRL key while you click each file to download it:

sales.csv

product_hierarchy.csv

store_cities.csv

store_names.csv

city_names.csv

product_names.csv

Software: MS Excel

Duration: 1 hr 30 minutes

Tasks:

1. Sales Data Cleaning

- Import Sales Data
  - Import the sales.csv files into Excel
- Explore the Data
  - Examine the number of rows and columns
  - Review the column names and their data types
  - Understand the significance of each column
- Identify Missing Values
  - Utilize filters (auto filters) to examine each column for missing values. Given that rows with missing data constitute a minor fraction of the data set, remove these rows to preserve data quality
- Format the Data
  - Date Formats: Standardize date formats utilizing Excel's date format tools
  - Numerical Consistency: Apply uniform number formats across your data set, including consistent use of decimal places and thousand separators
- Remove Duplicates
  - Use Excel's *Remove Duplicates* feature to find and eliminate any duplicate records within your data
- Standardize Text
  - Use the TRIM function to remove extra spaces, ensuring text data is uniformly formatted

2. Product Hierarchy Data Cleaning

- Import Product Hierarchy Data
  - Import the product_hierarchy.csv file into Excel and apply the same data cleaning steps
- Explore the Data
  - Examine the number of rows and columns
  - Review the column names and their data types
  - Understand the significance of each column
- Check for Missing Values
  - Identify and address any missing values in the data set.
- Format the Data
  - Ensure all product dimensions and hierarchies are correctly formatted and complete
- Remove Duplicates
  - Delete duplicate records from your data set
- Standardize Text
  - Apply text standardization techniques to ensure consistent data structure across the data set

3. Store Cities Data Cleaning

- Import Store Cities Data
  - Import the store_cities.csv file into Excel
- Explore the Data
  - Examine the number of rows and columns
  - Review the column names and their data types
  - Understand the significance of each column
- Perform Data Check
  - Identify Missing Values: Locate and correct missing values within the data set
  - Confirm Data Formatting: Verify that store information is consistent with sales data and that city details are accurate
  - Eliminate Duplicates: Remove duplicate entries from the data set
  - Standardize Text: Ensure textual data is consistent for analysis

4. Import Additional Data Files

- Import Supplementary Data
  - Import the product_names.csv, store_names.csv, and city_names.csv files into Excel
- Explore the Data
  - Examine the number of rows and columns
  - Review the column names and their data types
  - Understand the significance of each column
- Apply Data Cleaning Methodology
  - Utilize the established data cleaning process to prepare these additional data sets for subsequent analysis

5. Take Screenshots

- Take a screenshot of your final workbook
  - Ensure that all worksheet names are clearly visible
  - This screenshot should be included in your final presentation

Note

By following these structured steps, you ensure that all imported data sets are clean, consistent, and ready for further analysis or integration. This foundational work is crucial for accurate and insightful data analysis.

When applying data cleaning methodology to each data file (such as sales, product_hierarchy, store_cities), take screenshots of the ***before*** and ***after*** stages. For example, when you discover missing values, capture a screenshot of these, and then show another screenshot demonstrating that there are no missing values after removal. Similarly, check for duplicate values, take a screenshot of any duplicates, and then take another screenshot after they have been removed.

These screenshots should be included in your final presentation.

Congratulations! You have completed this lab and are ready for the next topic.

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